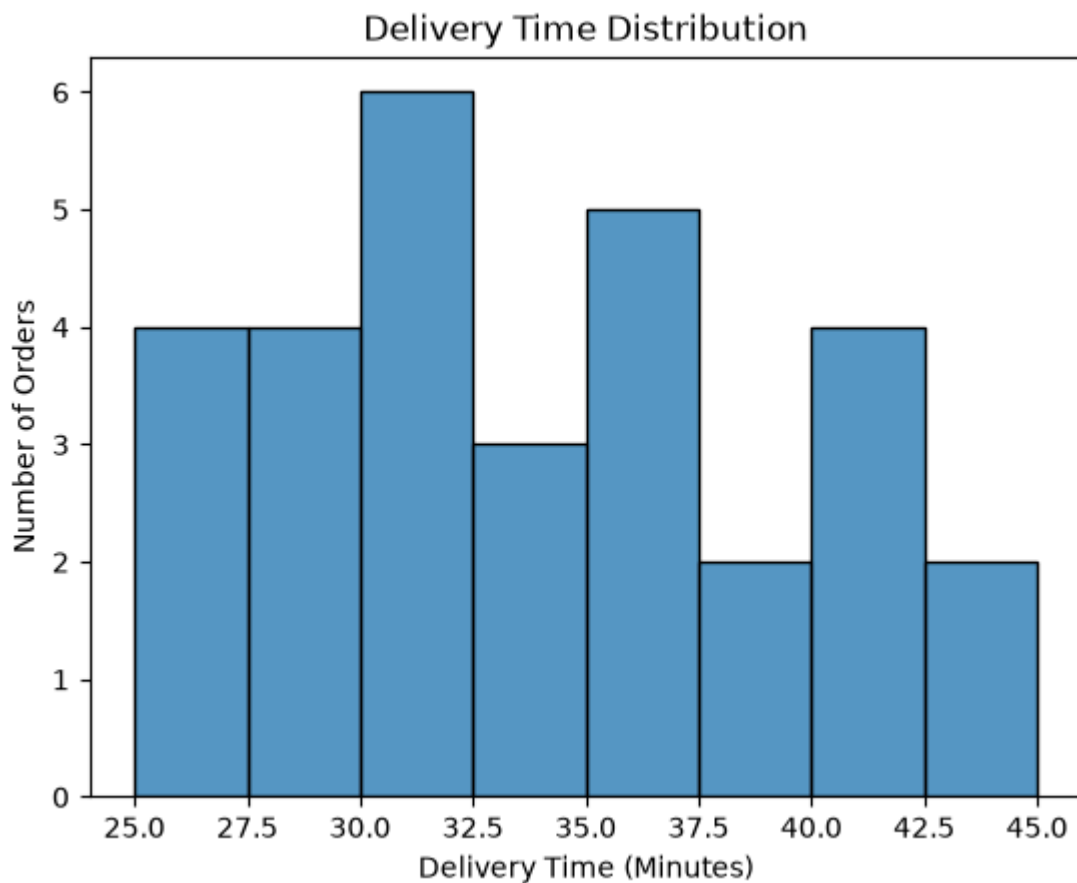


```
In [1]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: # TASK 1 - Delivery Time Distribution

delivery_time = [25, 30, 28, 35, 40, 32, 27, 45, 38, 30, 33, 29, 42, 36, 31, 28,

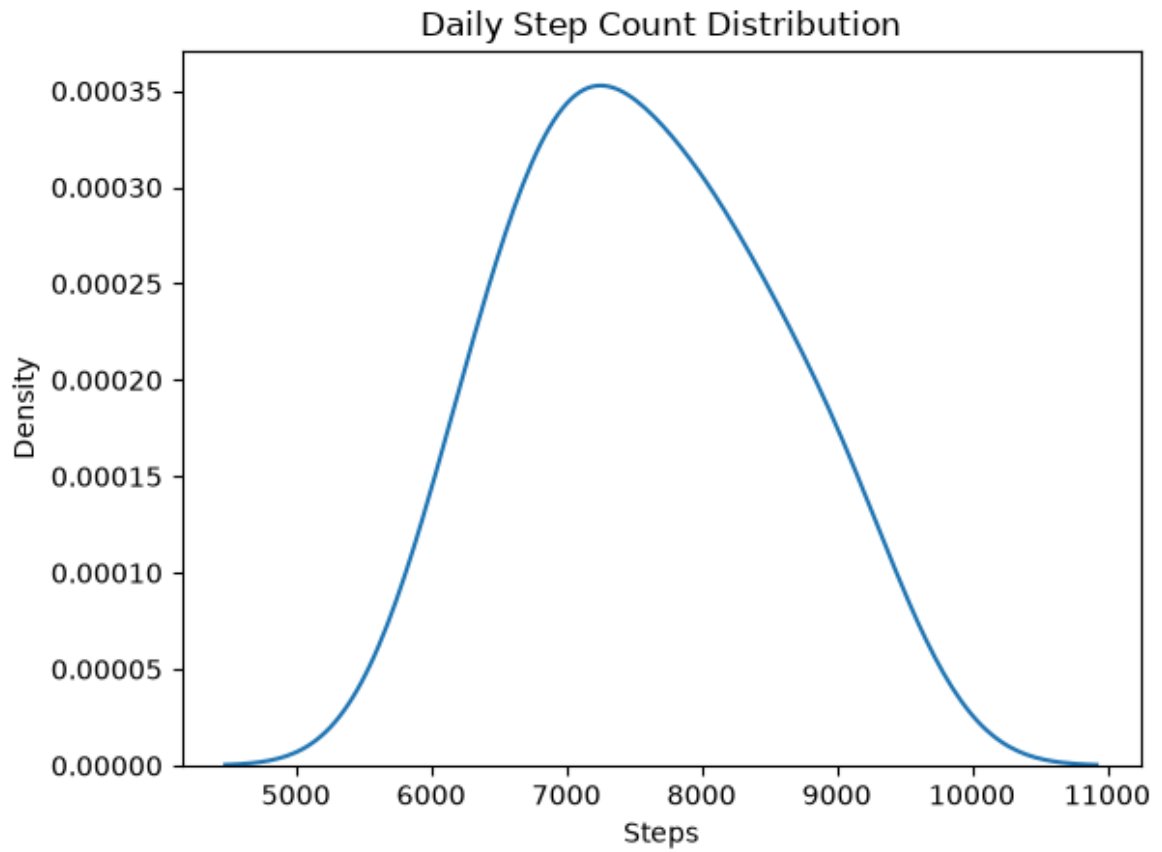
sns.histplot(delivery_time, bins=8)
plt.title("Delivery Time Distribution")
plt.xlabel("Delivery Time (Minutes)")
plt.ylabel("Number of Orders")
plt.show()
```



```
In [3]: # TASK 2 - Daily Steps for One Month

steps = [6500, 7200, 8100, 5900, 9000, 7500, 6800, 8200, 7100, 6400, 9500, 8800,
8500, 8000, 7400, 6300, 8900, 7700, 7200, 8300, 6800, 7900]

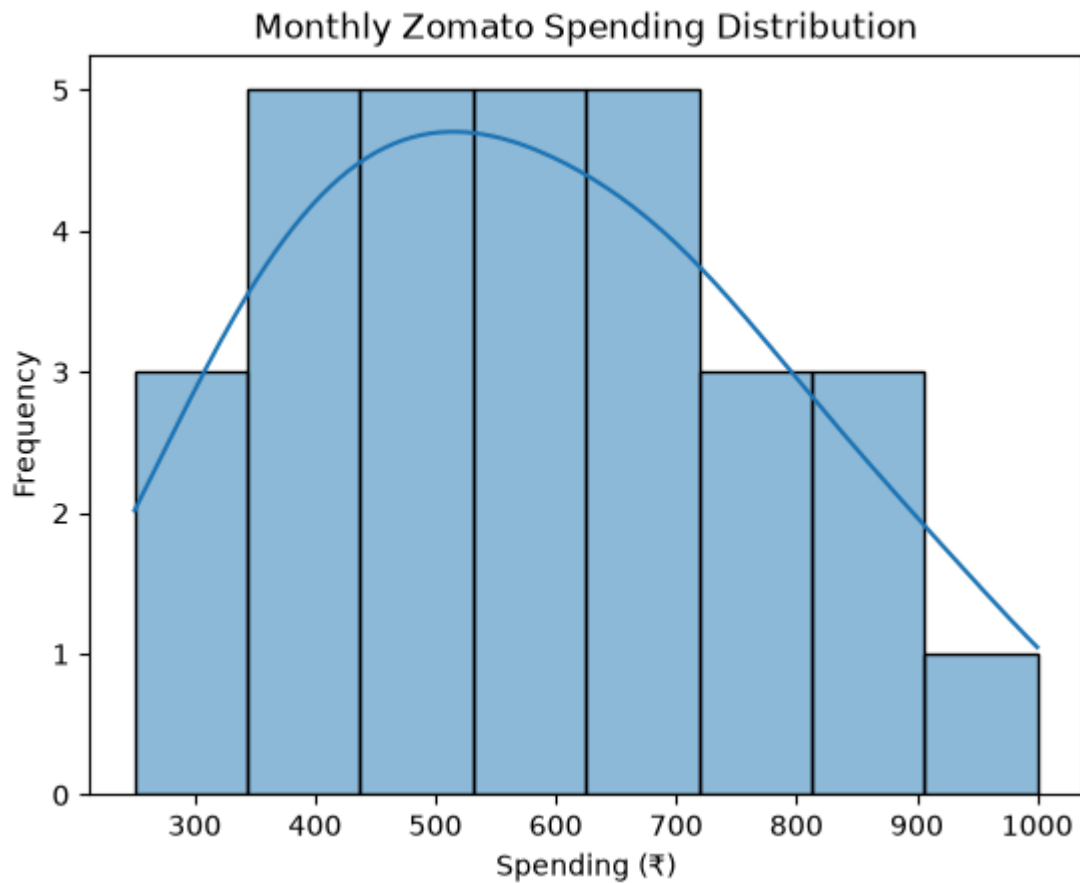
sns.kdeplot(steps)
plt.title("Daily Step Count Distribution")
plt.xlabel("Steps")
plt.show()
```



```
In [4]: # TASK 3 - Zomato Monthly Spending

spending = [250, 400, 550, 700, 350, 600, 800, 450, 500, 900, 300, 650, 750, 400,
350, 700, 500, 850, 450, 600, 300, 750, 650, 500, 900, 400, 700, 550]

sns.histplot(spending, bins=8, kde=True)
plt.title("Monthly Zomato Spending Distribution")
plt.xlabel("Spending (₹)")
plt.ylabel("Frequency")
plt.show()
```



In [5]: *# TASK 4 - IPL Run Distribution*

```
runs = [120, 145, 180, 165, 210, 190, 155, 175, 200, 185, 140, 220, 195, 160, 230,
135, 225, 175, 185, 210, 155, 200, 195, 165, 215]
```

```
# 5 bins
```

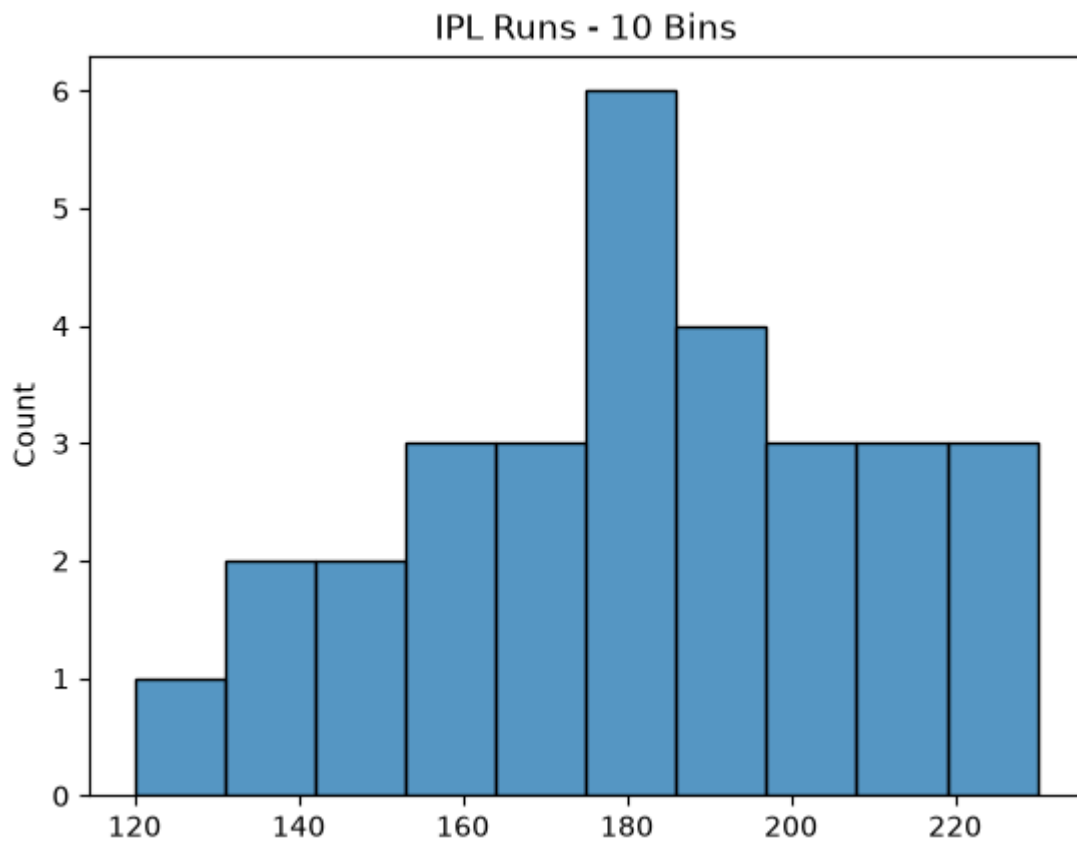
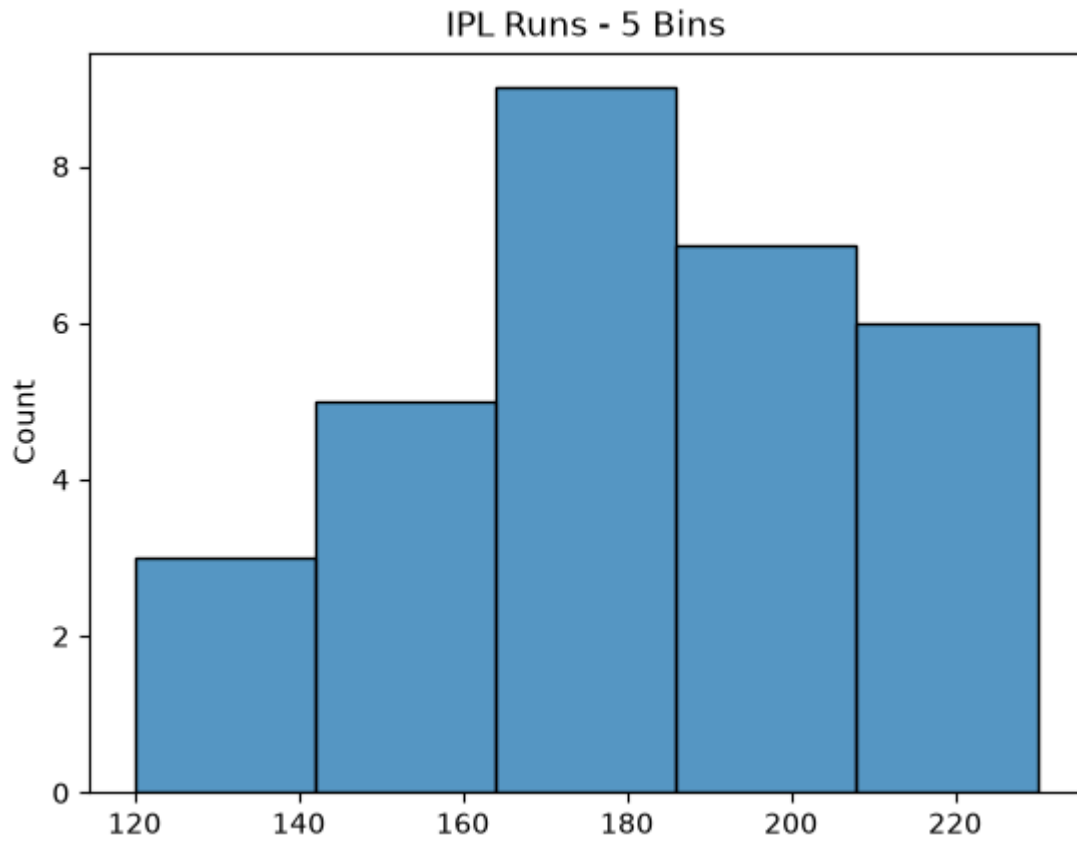
```
sns.histplot(runs, bins=5)
plt.title("IPL Runs - 5 Bins")
plt.show()
```

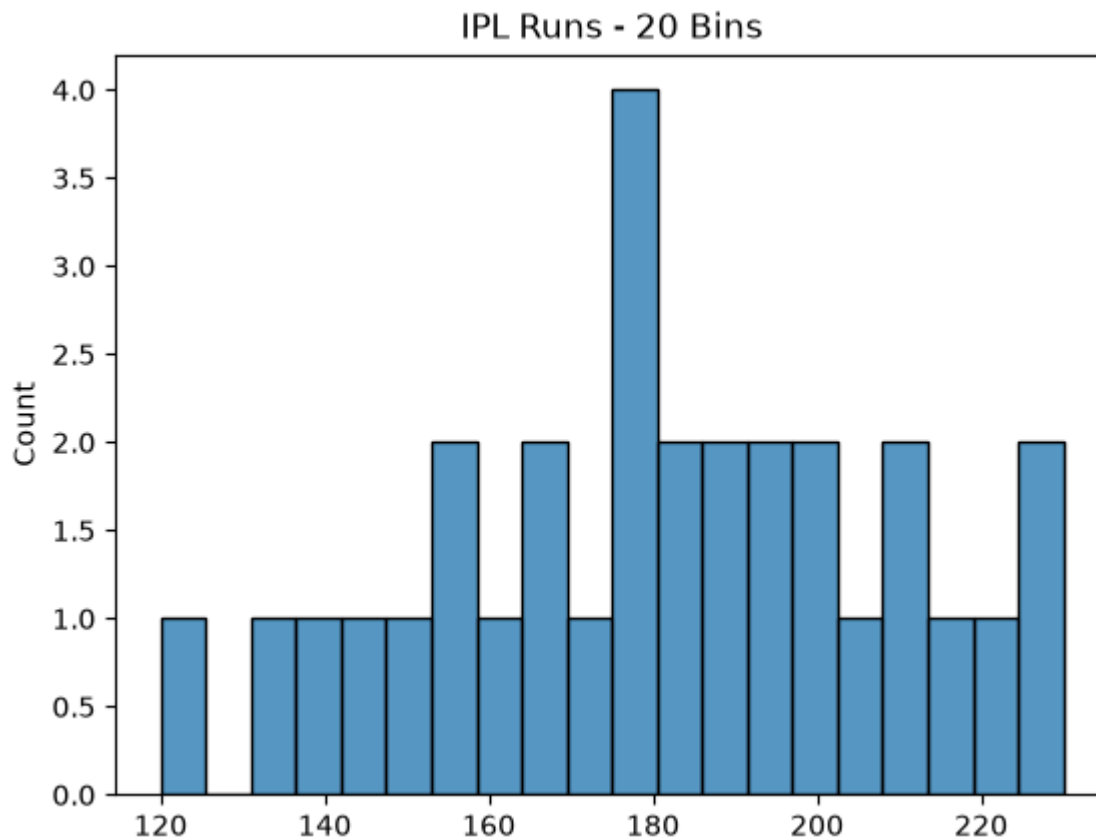
```
# 10 bins
```

```
sns.histplot(runs, bins=10)
plt.title("IPL Runs - 10 Bins")
plt.show()
```

```
# 20 bins
```

```
sns.histplot(runs, bins=20)
plt.title("IPL Runs - 20 Bins")
plt.show()
```





Analysis Task 1 – Delivery Time Distribution;- The distribution indicates that most orders are concentrated within the 25–40 minute range, suggesting relatively consistent delivery performance. Only a small number of orders fall outside this range.

Task 2 – Daily Step Count Distribution;- The KDE curve shows that daily activity is primarily concentrated around 7,000–8,000 steps. The relatively smooth distribution indicates that step counts remain fairly consistent throughout the month.

Task 3 – Zomato Spending Distribution;- The spending distribution is concentrated mainly between ₹400 and ₹700, indicating that most orders fall within this spending range. Higher-value orders occur less frequently, suggesting limited spending at the upper end.

Task 4 – IPL Run Distribution;- With 5 bins, the distribution provides a broad overview of scoring patterns. Increasing to 10 and 20 bins reveals greater variation and finer details in the run distribution, although excessive binning can make the pattern more difficult to interpret.